Project 6: Auto Research for Materials Discovery

Task split, handovers and timeline (6 members)

Who does what

One clear area per person. Each member builds their part, tests it, and prepares their own explanation and slide material. **Handover rule:** code goes to **M1**; experiment results go to **M2**; report text and slides go to **M6**.

Member	Builds and tests	Evidence	Handovers (what → who, by when)
M1: System integration	Installs MADE, builds the agent with an on/off switch for each part, merges everyone's code. Designs **budget management**: how the agent spends its query budget.	Runs the full frozen agent once (the reference every ablation compares against). Reproducibility package.	Code: receives from M2–M5 by 24 Oct. Reference results → M2 by 26 Oct noon. Text and slides → M6 (method 24 Oct, results 27 Oct).
M2: Baselines and evaluation	Runs Random and Chemeleon baselines; scoring (mSUN@B, AUDC, AF/EF); discovery curves. Designs **tool routing** (which generator, filter or scorer to use). Backup for M1 during setup.	Tool routing ablation. Final results table: us vs. both baselines. Cost analysis.	Receives all results by 27 Oct. Tool routing code → M1 by 24 Oct. Results table and curves → M6 by 27 Oct.
M3: Planning and self-correction	How the agent breaks the goal into steps, plus detecting when it's stuck and re-planning.	Two ablations: planning off; self-correction off.	Code → M1 by 24 Oct. Results → M2 by 27 Oct. Text and slides → M6 (method 24 Oct, results 27 Oct).
M4: Memory	How the agent records past attempts and reuses them; keeps the evidence log. Picks example discoveries.	Memory ablation.	Code → M1 by 24 Oct. Logs → M5 from first run. Results → M2 by 27 Oct. Text and slides → M6 (method 24 Oct, results 27 Oct).
M5: Exploration	Balances searching new areas vs. digging deeper; finds the 2 failure examples (over-exploitation, wasteful exploration).	Exploration ablation plus failure trajectories.	Code → M1 by 24 Oct. Results → M2 by 27 Oct. Text and slides → M6 (method 24 Oct, results 27 Oct).
M6: Experiment design, generalization, report lead	Designs the validation protocol (tuning vs. check split of practice materials; seeds; budgets). Runs the hidden-test comparison.	Practice vs. hidden-test comparison with per-material breakdown; combines all ablations.	Receives text and slides (24 and 27 Oct). Hidden-test results → M2 by 27 Oct. Final report and slides by 29 Oct.

*Not in scope: optional extensions such as multi-agent or self-improvement are left as **future work**; each member can mention them in their 'Future Plans' section.*

Timeline

Date	What happens
17–18 Oct	Everyone: install MADE, time one run, kickoff call, start work log. M6: Group C sheet; questions to TA.
19–20 Oct	M1: agent skeleton running (**checkpoint 20 Oct**). M2: baselines and scoring. M3–M5: design their part. M6: validation protocol and report template.
21–24 Oct	M2–M5 build and test their parts on practice materials. M6 runs check-set tests.
24 Oct	**Handover 1:** code → M1; method text and draft slides → M6.
25 Oct	**Freeze the agent** (M1 tags the version).
26 Oct	M1 runs the reference (results → M2 by noon). Everyone starts their own ablation, on their own laptop, with the smaller ablation budget. M6 starts hidden-test runs.
27 Oct	**Handover 2:** all experiment results → M2; results text and slides → M6. M2 sends the final table and curves → M6.
28 Oct	M6 assembles the report and slides; members review their own sections.
29 Oct	Full rehearsal (10 min + mock Q&A); final edits.
30 Oct	Submission deadline (confirm with TA what is due).
31 Oct / 1 Nov	Presentation: everyone presents their own part and answers Q&A.
3 Nov	Individual reports due (max 3 pages, to YAOLONG001@e.ntu.edu.sg).

Team rules

- **Keep a work log:** what you built, what failed, what you learned. It's the raw material for your individual report.
- **Negative results are fine:** explain why your part didn't help and what you'd change.
- **Explain the why:** e.g. 'memory raised mSUN by 12% because the agent stopped re-testing failed materials'.
- **Run times differ by laptop:** label cost figures with the machine used.